

Midwife Copilot – Prototype Concept and Technical Development Status

Technical Report | Working status as of 29 September 2026

Purpose of this document: This report describes the functional and technical concept of the current Midwife Copilot prototype, clearly separates already implemented functionality from the planned next development steps, and outlines the intended Phase 2 and target architecture. The documented system is a pilot and development prototype, not yet a production-ready clinical system.

1. Product concept and core principle

Midwife Copilot is an AI-supported documentation assistant for hospital-based midwives in maternity care. The goal is to move documentation closer to the care process: observations can be captured during or after care by voice or text. Voice input is first converted into a transcript; relevant information is then extracted and mapped into a structured documentation draft. The midwife reviews and edits the result, commits it to the record, and only then confirms the documentation through sign-off.

Voice / Text → Transcript → AI Structuring → Review · Sign-off · Export

A central design principle is human-in-the-loop control. The AI only produces an editable documentation draft; it does not diagnose, recommend treatment, or autonomously finalise a clinical record. Transcript, structured fields, and chronological course information remain reviewable by the midwife, and responsibility for the final record remains with the healthcare professional.

2. Current prototype – already implemented

Phase 1 was implemented as a browser-based web application and tested with fictitious or fully anonymised cases. The focus was deliberately limited to the core documentation workflow and to validating whether voice capture, speech-to-text, AI-based structuring, review, and sign-off can work in the delivery-room context. The pilot confirmed the basic approach while identifying concrete requirements around time logic, overview, clinical flexibility, and technical reliability.

Component	Current implementation	Status
Capture	Voice input and free-text input in the web application; usable on desktop and in a mobile browser.	implemented
Speech-to-text	Server-side transcription followed by manual correction. ElevenLabs was used in the Phase 1 setup; alternative STT options are being evaluated.	implemented / replaceable

MIDWIFE COPILOT – TECHNICAL REPORT (DRAFT)

Component	Current implementation	Status
AI structuring	Conversion of corrected text into structured clinical entries. Mistral was used for structuring in the pilot.	implemented
Review	Structured fields can be reviewed and edited before sign-off; missing information and uncertain extractions can be surfaced for checking.	implemented
Time & course view	Spoken event times are taken into account, and a time-based/chronological documentation view has been added.	implemented, still being refined
Traceability	Structured information can be linked back to the dictated source; uncertain values can be highlighted.	implemented
Sign-off & export	Sign-off is represented in the prototype; confirmed documentation can be exported as PDF.	implemented
Warning logic	The user is warned before sign-off when drafts or recordings have not yet been incorporated into the record.	implemented

Around 30 changes have already been incorporated into a revised prototype version based on Phase 1 feedback. The most visible improvements include substantially shorter AI processing time (from approximately 40–55 seconds observed during testing to around 7 seconds), an improved mobile dictation flow, clearer error messages, better separation of recording time and clinical event time, warnings for open drafts, and a chronological overall view.

3. Technical concept of the current prototype

The prototype is built as a lean server-side processing pipeline. The user interface runs in the browser; audio and text are sent to a backend that orchestrates the processing steps. AI processing is not performed on the end device. This makes it possible to exchange speech-to-text and language-model components independently and to compare them based on accuracy, cost, privacy, and regulatory suitability.

1. Midwife Web App	Browser interface for case selection, voice/text capture, transcript correction, structuring, review, sign-off, and PDF output.
2. Backend / API	Receives inputs, orchestrates processing, and connects the user interface, AI services, and persistent case data.
3. Speech-to-Text	Converts audio into text. The component is intentionally replaceable; reliable recognition of German obstetric language is a key selection criterion.

MIDWIFE COPILOT – TECHNICAL REPORT (DRAFT)

1. Midwife Web App	Browser interface for case selection, voice/text capture, transcript correction, structuring, review, sign-off, and PDF output.
4. LLM Structuring	Extracts relevant information from the transcript and maps it to the structured documentation schema without inventing missing clinical facts.
5. Documentation Model	Combines time-keyed structured entries with free text and contextual observations. Milestones, chronology, and categories are different views of the same case.
6. Review & Output	All entries remain editable before sign-off. After confirmation, the record can be exported; direct HIS/KIS transfer is part of the target architecture, not the current core prototype.

3.1 Documentation logic

The target documentation schema includes clinical key data such as maternal vital signs, fetal status/CTG, vaginal examination findings, interventions, medication, complications, mode of birth, neonatal parameters, and free-text observations. Phase 1 showed clearly that birth documentation cannot be represented solely through rigid structured fields. The prototype therefore combines structured fields for key facts with a chronological course view and free text for context and clinically relevant side observations.

A further key requirement is the distinction between documentation time and event time. Midwives frequently document retrospectively. Therefore, an entry such as “23:45 CTG applied” must be sorted at 23:45 even when the dictation is recorded later. This logic has already been improved, but robust handling of several events or several time references within one dictation remains an active development area.

3.2 Current technical boundary

The current environment is a pilot setup intended for anonymised test data. It is not yet approved as a production environment for real patient data. The cloud and backend setup is currently being hardened. In particular, separate development/test and production environments, private network and database access, secrets management, backup/restore, monitoring, and controlled deployment and rollback processes are prerequisites for the planned real-world Phase 2.

4. Planned next prototype development

The next development steps focus less on changing the core workflow and more on making it clinically robust and production-oriented. Based on the current project plan, prototype development is expected to continue for approximately three further weeks after the end of September, with iterative testing and direct incorporation of clinical feedback.

MIDWIFE COPILOT – TECHNICAL REPORT (DRAFT)

Area	Planned / to be finalised	Expected horizon
Time logic	Recognise several events and time points within one dictation and reliably place retrospective additions in chronological order.	short term
Documentation views	Further simplify overview and navigation; align milestones, chronological view, and category view consistently.	short term
Case-specific logic	Different requirements for outpatient/pre-admission cases, inpatient birth, and caesarean section; context-sensitive mandatory fields.	short term
Clinical detail	Expand CTG/FIGO, vaginal examination/cervix/Portio, rupture of membranes, newborn/U1, birth injuries, emergencies, medication, allergies, and risks.	short term
Mobile UX & stability	Improve smartphone/tablet layout, recording reliability, loading states, error handling, German UI, and 24-hour time display.	short term
Amendments	Controlled post-sign-off amendment process with versioning rather than overwrite.	next expansion stage
AI models	Continue comparing STT and LLM options, including Mistral/Corti and suitable hyperscaler-based alternatives; later potentially domain-specific tuning.	ongoing / post-pilot

5. Target architecture for Phase 2 and later clinical integration

Phase 2 is intended to move from feasibility validation with anonymised cases to a more realistic pilot using pseudonymised real birth cases. The current pilot architecture is not sufficient for this step. The functional logic remains largely the same, but identity, security, governance, and integration components need to be added.

Target component	Intended function
Midwife App	Remains the primary working surface for capture, review, and sign-off; browser-based and potentially later complemented by native mobile applications.
HPMP – Hospital Patient Management Portal	Separate clinic-admin surface for configuration, users, roles/access, patient assignment, tenant separation, audit functions, and HIS/KIS handoff. Conceptually specified, but not yet implemented as a complete pilot admin product.
Authentication & roles	Production-oriented login with username/password + 2FA, personal sign-off PIN, and role-based permissions. The role/access model is still being finalised.
Security hardening	Separate dev/test/prod environments, private VPC/database architecture, secrets management, encryption, controlled deployment/rollback, backups with restore testing, monitoring/logging/alerting.

MIDWIFE COPILOT – TECHNICAL REPORT (DRAFT)

Target component	Intended function
Privacy & governance	DPA/AVV, DPIA, defined retention/deletion concepts, technical and organisational measures, and use of suitable service providers before processing real patient data.
HIS/KIS integration	Short term: handoff of the confirmed document as PDF. Later: structured exchange via HL7/FHIR, with future alignment to systems such as ORBIS/ORBIS Storch.

5.1 Security and compliance path

For Phase 2, the intended environment is a hardened setup oriented toward C5 readiness. This does not mean that the current prototype is already C5-compliant or certified. Before processing real cases, the relevant hardening measures should be in place, the DPIA and data-processing arrangements completed, and third-party service providers assessed. A later independent C5 attestation is part of the broader clinical-readiness target.

5.2 Deliberate product boundaries

Documentation assistance, not clinical decision support: no diagnosis, treatment recommendation, or autonomous risk assessment.

No autonomous sign-off: every AI output remains reviewable and editable; human sign-off is the final gate.

No production processing of real patient data in the Phase 1 prototype.

No complete HIS/KIS integration in the current prototype; PDF export is the current handoff mechanism.

Not part of the current core prototype: a fully implemented HPMP, a separate neonatal workflow, multimodal handwriting/photo capture, a fine-tuned proprietary speech model, and outpatient/freelance midwifery workflows.

6. Summary

The current Midwife Copilot is a functional pilot prototype that supports the end-to-end core workflow “Dictate/Text → Transcribe → AI Structure → Review → Sign-off → PDF”. Phase 1 indicates that voice input, transcription quality, and AI-supported structuring are technically feasible. The main remaining work concerns robust time and amendment logic, a more flexible clinical documentation structure, mobile reliability, additional clinical detail, and security/governance hardening.

The target architecture is deliberately modular. The tested web application and AI pipeline are intended to be extended rather than replaced: additional admin and role functions, auditability, privacy-compliant processing of real cases, and HIS/KIS integration can be added step by step. This allows the existing prototype to serve as the technical basis for Phase 2 validation and, subsequently, for a clinically integrated product.